American Computer Science League

2021-2022 ● Contest 2: Shorts ● Classroom DivisionSolutions

1. Prefix/Infix/Postfix

+ / 9 3 + * 4 − 8 3 ^ 5 2 = + (/ 9 3) + * 4 (− 8 3) (^ 5 2)
= + 3 + (* 4 5) 25
= + 3 (+ 20 25)
= (+ 3 45)
= 48

C. 48

2. Prefix/Infix/Postfix

$$\frac{(A^2 + B)(B^2 - C)}{B + D^2}$$

= ((A² + B) * (B² − C)) / (B + D²)
= ((A 2 ^ B +) * (B 2 ^ C −)) / (B D 2 ^ +)
= (A 2 ^ B + B 2 ^ C −*) / (B D 2 ^ +)
= A 2 ^ B + B 2 ^ C − * B D 2 ^ + /

D. A 2 ^ B + B 2 ^ C − * B D 2 ^ + /

3. Bit-String Flicking

01100 OR 11011 AND 01010 = 01100 OR (11011 AND 01010)
= 01100 OR 01010
= 01110

B. 01110

4. Bit-String Flicking

(LCIRC-2 01100) AND NOT 11010 OR (RSHIFT-1 01100)
= 10001 AND 00101 OR 00110
= 00001 OR 00110
= 00111

C. 00111

5. LISP

(CAADDR '((1 2 (3)) (4 (5) 6) ((5 4) (3 (2)))))
= (CAADR '((4 (5) 6) ((5 4) (3 (2)))))
= (CAAR '(((5 4) (3 (2)))))
= (CAR '((5 4) (3 (2))))
= (5 4)

D. (5 4)

6. Prefix/Infix/Postfix

6 2 ^ 6 − 5 / 8 6 2 + / − 5 / = (6 2 ^) 6 − 5 / 8 (6 2 +) / − 5 /
= (36 6 −) 5 / (8 8 /) − 5 /
= (30 5 /) 1 − 5 /
= (6 1 −) 5 /
= (5 5 /)
= 1

B. 1

7. Prefix/Infix/Postfix

/ * + − 5 4 5 ^ 1 4 / 9 3 = / * + (− 5 4) 5 (^ 1 4) (/ 9 3)
= / * (+ (5 4 −) 5) (1 4 ^) (9 3 /)
= / (* (5 4 − 5 +) (1 4 ^)) (9 3 /)
= (/ (5 4 − 5 + 1 4 ^ *) (9 3 /))
= 5 4 − 5 + 1 4 ^ * 9 3 / /

E. 5 4 − 5 + 1 4 ^ * 9 3 / /

8. Bit-String Flicking

00100 OR 10100 AND NOT (01111 OR 00101)
= 00100 OR 10100 AND NOT 01111
= 00100 OR (10100 AND 10000)
= 00100 OR 10000
= 10100

A. 10100

9. Bit-String Flicking (LCIRC-2 (10011 AND (NOT 11101) OR (RSHIFT-2 11010))) = (LCIRC-2 (10011 AND 00010 OR 00110)) = (LCIRC-2 (00010 OR 00110)) = (LCIRC-2 00110) = 11000	C. 11000
10. LISP (CAR (CDR '((0 1) (3 2 1) (2 4 1) (3 2)))) = (CAR '((3 2 1) (2 4 1) (3 2))) = (3 2 1)	C. (3 2 1)